

JEE MAIN 2026 — HARDCORE FULL-LENGTH MOCK TEST

Advanced-Inspired • Original Questions • 75 Questions • 300 Marks • 3 Hours

EXAM INSTRUCTIONS

- Paper 1 (B.E./B.Tech.) style: Physics, Chemistry and Mathematics.
- Each subject contains 20 single-correct MCQs and 5 numerical-value questions: 25 questions per subject, 75 questions total.
- Maximum marks: 300. For MCQs and numerical-value questions: +4 for a correct response, –1 for an incorrect response, 0 for unanswered.
- Numerical answers should be entered as a non-negative integer/appropriate numerical value as instructed by the actual computer-based examination. For this paper, report the value requested in each question.
- Use $g = 10 \text{ m/s}^2$ unless otherwise stated. Use standard atomic masses and constants where needed.
- All questions in this mock are newly formulated. They are not claimed to be official or previous-year JEE questions.
- The question paper follows the current JEE Main Paper 1 framework; the difficulty is deliberately raised with Advanced-inspired conceptual combinations.

Pattern reference: NTA JEE(Main) 2026 Information Bulletin and official 2026 JEE sources.

PHYSICS

Q1. [MCQ] A particle moves in a straight line with $x(t)=t^3-6t^2+9t$ (SI). The ratio of the total distance travelled to the magnitude of displacement during $0 \leq t \leq 4$ s is:

- A) 1
- B) 2
- C) 3
- D) 4

Q2. [MCQ] A block of mass m rests on a rough wedge of angle θ . The wedge is accelerated horizontally so that the block remains at rest relative to it. Neglecting limiting friction, the required horizontal acceleration magnitude is:

- A) $g \tan \theta$
- B) $g \cot \theta$
- C) $g \sin \theta$
- D) $g/\cos \theta$

Q3. [MCQ] A uniform rod of length L and mass M is hinged at one end and released from rest from the horizontal. Its angular speed when it reaches the vertical position is:

- A) $\sqrt{g/L}$
- B) $\sqrt{2g/L}$
- C) $\sqrt{3g/L}$
- D) $\sqrt{6g/L}$

Q4. [MCQ] A small block slides from rest down a smooth quarter-circular track of radius R and then enters a rough horizontal surface of coefficient μ . The minimum length of rough surface required to stop it is:

- A) R/μ
- B) $2R/\mu$
- C) $R/(2\mu)$
- D) μR

Q5. [MCQ] A 2 kg block on a smooth table is connected to a 1 kg hanging mass. A constant 4 N force acts horizontally on the 2 kg block opposite its motion. Take $g=10 \text{ m/s}^2$. The acceleration is:

- A) 1 m/s^2
- B) 2 m/s^2
- C) 3 m/s^2
- D) 4 m/s^2

Q6. [MCQ] A satellite moves in a circular orbit of radius r . If its orbital radius is suddenly changed to $4r$ while its angular momentum about the planet remains unchanged, its new speed is:

- A) $v/4$
- B) $v/2$
- C) $2v$
- D) $4v$

Q7. [MCQ] A liquid rises to height h in a capillary of radius r . If the same capillary is used on a planet where surface gravity is $g/4$, with all other properties unchanged, the rise is:

- A) $h/4$
- B) $h/2$
- C) $2h$
- D) $4h$

Q8. [MCQ] One mole of an ideal gas undergoes a reversible process $P \propto V^2$ from V to $2V$. If initial pressure is P , the work done by the gas is:

- A) PV
- B) $2PV$
- C) $7PV/3$

D) 3PV

Q9. [MCQ] A particle executes SHM $x=A \cos \omega t$. The ratio of kinetic energy to potential energy when $x=A/2$ is:

- A) $1/3$
- B) 1
- C) 3
- D) 4

Q10. [MCQ] A string fixed at both ends has fundamental frequency f . If tension is increased by 44% and length is decreased by 20%, the new fundamental frequency is:

- A) $1.2f$
- B) $1.5f$
- C) $1.8f$
- D) $2f$

Q11. [MCQ] A point charge q is placed at the centre of a conducting spherical shell carrying net charge Q . The charge induced on the inner surface is:

- A) 0
- B) $+q$
- C) $-q$
- D) $Q-q$

Q12. [MCQ] Two capacitors C and $2C$ are charged separately to the same voltage V and then connected positive-to-positive and negative-to-negative. The final common voltage is:

- A) $V/2$
- B) $2V/3$
- C) V
- D) $3V/2$

Q13. [MCQ] A cell of emf E and internal resistance r supplies maximum power to an external resistor R . The efficiency at maximum power transfer is:

- A) 25%
- B) 50%
- C) 75%
- D) 100%

Q14. [MCQ] A charged particle enters a uniform magnetic field perpendicular to its velocity. If its kinetic energy is doubled while charge and field remain unchanged, the radius becomes:

- A) $r/2$
- B) $r/\sqrt{2}$
- C) $\sqrt{2}r$
- D) $2r$

Q15. [MCQ] A conducting rod of length l rotates with angular speed ω about one end in a uniform magnetic field B perpendicular to the plane of rotation. The emf between its ends is:

- A) $Bl\omega$
- B) $B\omega l^2$
- C) $\frac{1}{2}B\omega l^2$
- D) $2B\omega l^2$

Q16. [MCQ] In a series LCR circuit at resonance, the current is I . If the frequency is doubled while R, L, C remain fixed, which statement is necessarily correct?

- A) Current doubles
- B) Current becomes zero
- C) Impedance becomes R
- D) Impedance becomes greater than R

Q17. [MCQ] An object is placed 30 cm in front of a convex lens of focal length 20 cm. A plane mirror is placed 10 cm behind the lens. The final image after reflection and second passage through the lens is formed at:

- A) 20 cm in front of lens
- B) 30 cm in front of lens
- C) 40 cm in front of lens
- D) at infinity

Q18. [MCQ] In Young's double-slit experiment, the intensity at a point is I_0 when the phase difference is $\pi/3$. If the intensities from the two slits are equal, the maximum intensity is:

- A) $2I_0$
- B) $4I_0/3$
- C) $3I_0/2$
- D) $4I_0$

Q19. [MCQ] A photon of wavelength λ ejects electrons from a metal with stopping potential V_0 . If the wavelength is changed to $\lambda/2$, the new stopping potential is:

- A) V_0
- B) $2V_0$
- C) $2V_0 + hc/(e\lambda)$
- D) $V_0 + hc/(e\lambda)$

Q20. [MCQ] A nucleus has mass defect Δm . If its binding energy per nucleon is 7.5 MeV and it contains 240 nucleons, Δm in atomic mass units is approximately:

- A) 0.75 u
- B) 1.93 u
- C) 3.00 u
- D) 7.5 u

Q21. [NUM] A particle is projected vertically upward with speed 20 m/s. Take $g=10 \text{ m/s}^2$. Find the total distance travelled in the first 3 s.

Enter the required numerical value.

Q22. [NUM] A 2 kg block is pulled on a horizontal surface by a force $F=10+2x$ N from $x=0$ to $x=5$ m. Friction is constant at 4 N. If it starts from rest, find its speed at $x=5$ m (take g irrelevant).

Enter the required numerical value.

Q23. [NUM] A capacitor of capacitance $6 \mu\text{F}$ is charged to 100 V and then disconnected. A dielectric of $K=3$ completely fills the gap. Find the final stored energy in mJ.

Enter the required numerical value.

Q24. [NUM] A first-order reaction has half-life 20 min. How much time is required for 87.5% of the reactant to disappear?

Enter the required numerical value.

Q25. [NUM] For the quadratic $x^2 - 6x + k = 0$, the roots differ by 2. Find k .

Enter the required numerical value.

CHEMISTRY

- Q1. [MCQ] A 0.10 mol sample of an ideal gas expands isothermally at 300 K from 2 L to 8 L. Work done by the gas is closest to:
- A) 0.35 kJ
 - B) 0.69 kJ
 - C) 1.04 kJ
 - D) 1.38 kJ
- Q2. [MCQ] For a buffer containing equal concentrations of CH_3COOH and CH_3COO^- , the pH is:
- A) $\text{pK}_a - 1$
 - B) pK_a
 - C) 7
 - D) $14 - \text{pK}_a$
- Q3. [MCQ] For the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, increasing pressure at constant temperature shifts equilibrium:
- A) right
 - B) left
 - C) no shift
 - D) first right then left
- Q4. [MCQ] A galvanic cell has $E^\circ_{\text{cathode}} = +0.80 \text{ V}$ and $E^\circ_{\text{anode}} = -0.76 \text{ V}$. Its standard EMF is:
- A) 0.04 V
 - B) 0.76 V
 - C) 1.56 V
 - D) -1.56 V
- Q5. [MCQ] The rate constant of a first-order reaction is $2.31 \times 10^{-3} \text{ s}^{-1}$. Its half-life is approximately:
- A) 150 s
 - B) 300 s
 - C) 450 s
 - D) 600 s
- Q6. [MCQ] Which species has the greatest bond order according to MO theory?
- A) O_2
 - B) O_2^+
 - C) O_2^-
 - D) O_2^{2-}
- Q7. [MCQ] A compound gives a positive iodoform test and also reacts with Na metal to liberate H_2 . Which is consistent?
- A) CH_3COCH_3
 - B) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
 - C) $\text{C}_6\text{H}_5\text{CHO}$
 - D) CH_3OCH_3
- Q8. [MCQ] The major product of dehydration of 2-methyl-2-butanol under acidic conditions is:
- A) 2-methyl-1-butene
 - B) 2-methyl-2-butene
 - C) 3-methyl-1-butene
 - D) but-2-ene
- Q9. [MCQ] Which reagent selectively distinguishes phenol from ethanol?
- A) Na metal
 - B) neutral FeCl_3
 - C) H_2SO_4

D) PCl_5

Q10. [MCQ] For an $\text{S}_\text{N}1$ reaction, the rate-determining step is:

- A) nucleophile attack
- B) carbocation formation
- C) proton transfer
- D) product deprotonation

Q11. [MCQ] A 0.1 m solution of a non-electrolyte has freezing point depression 0.186°C . Taking $K_f = 1.86 \text{ K kg mol}^{-1}$, the van't Hoff factor is:

- A) 0.1
- B) 0.5
- C) 1
- D) 2

Q12. [MCQ] In a face-centered cubic crystal, the number of atoms effectively present per unit cell is:

- A) 2
- B) 4
- C) 6
- D) 8

Q13. [MCQ] The oxidation state of Fe in $[\text{Fe}(\text{CN})_6]^{4-}$ is:

- A) +1
- B) +2
- C) +3
- D) +4

Q14. [MCQ] Which has the highest boiling point among the following, primarily due to hydrogen bonding?

- A) CH_3OCH_3
- B) $\text{CH}_3\text{CH}_2\text{OH}$
- C) CH_3CHO
- D) CH_3COCH_3

Q15. [MCQ] The number of unpaired electrons in Mn^{2+} ($Z=25$) is:

- A) 1
- B) 3
- C) 5
- D) 7

Q16. [MCQ] Which oxide is amphoteric?

- A) Na_2O
- B) MgO
- C) Al_2O_3
- D) SO_3

Q17. [MCQ] For a reaction $\text{A} + \text{B} \rightarrow \text{products}$, doubling $[\text{A}]$ doubles rate while doubling $[\text{B}]$ leaves rate unchanged. The overall order is:

- A) 0
- B) 1
- C) 2
- D) 3

Q18. [MCQ] Which pair can show geometrical isomerism?

- A) but-1-ene
- B) but-2-ene
- C) 2-methylpropene
- D) propene

Q19. [MCQ] In qualitative analysis, AgCl dissolves in aqueous NH_3 mainly because:

- A) AgCl is reduced
- B) Ag⁺ forms a soluble complex
- C) Cl⁻ is oxidized
- D) NH₃ precipitates Ag⁺

Q20. [MCQ] Which statement about an ideal gas is correct?

- A) molecules attract strongly
- B) internal energy depends only on pressure
- C) $PV=nRT$
- D) collisions are inelastic

Q21. [NUM] For 2 mol of an ideal gas, $R=8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, find the increase in internal energy for a monatomic gas heated by 50 K.

Enter the required numerical value.

Q22. [NUM] A 0.50 M weak acid HA has $K_a=1.0 \times 10^{-5}$. Assuming $x \ll C$, find $[H^+] \times 10^3$.

Enter the required numerical value.

Q23. [NUM] A first-order reaction is 75% complete in 20 min. Find its half-life in minutes.

Enter the required numerical value.

Q24. [NUM] Find the pH of 0.001 M HCl at 25°C.

Enter the required numerical value.

Q25. [NUM] A complex $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ contains how many chloride ions outside the coordination sphere per formula unit?

Enter the required numerical value.

MATHEMATICS

Q1. [MCQ] If $z=(1+i)/(1-i)$, then z^8 equals:

- A) 1
- B) -1
- C) i
- D) $-i$

Q2. [MCQ] For real x , the minimum value of x^2+4/x^2 is:

- A) 2
- B) 4
- C) 6
- D) 8

Q3. [MCQ] The number of real roots of $|x-1|+|x+1|=4$ is:

- A) 0
- B) 1
- C) 2
- D) 4

Q4. [MCQ] If A is a 2×2 matrix with $\det A=3$, then $\det(2A)$ is:

- A) 6
- B) 9
- C) 12
- D) 18

Q5. [MCQ] The coefficient of x^3 in $(1+x)^4(1-x)^4$ is:

- A) -8
- B) 0
- C) 8
- D) 16

Q6. [MCQ] The equation of the tangent to $x^2+y^2=25$ at $(3,4)$ is:

- A) $3x+4y=25$
- B) $4x+3y=25$
- C) $3x-4y=25$
- D) $x+y=7$

Q7. [MCQ] The eccentricity of the ellipse $x^2/25+y^2/9=1$ is:

- A) $3/5$
- B) $4/5$
- C) $5/4$
- D) $2/5$

Q8. [MCQ] $\lim_{x \rightarrow 0} [\sin 3x - 3\sin x]/x^3$ equals:

- A) -4
- B) -3
- C) 0
- D) 4

Q9. [MCQ] If $f(x)=|x|$, then f is:

- A) differentiable everywhere
- B) continuous but not differentiable at 0
- C) discontinuous at 0
- D) neither continuous nor differentiable

Q10. [MCQ] The maximum value of $x(4-x)$ for $x \in [0,4]$ is:

- A) 2
- B) 4
- C) 6
- D) 8

Q11. [MCQ] $\int_0^1 \frac{dx}{(1+x^2)}$ equals:

- A) $\pi/6$
- B) $\pi/4$
- C) $\pi/3$
- D) 1

Q12. [MCQ] If $\int_0^\pi \sin^n x \, dx = \int_0^\pi \sin^{(n+2)} x \, dx$, then n equals:

- A) 0
- B) 1
- C) -1
- D) 2

Q13. [MCQ] The general solution of $dy/dx=2y$ is:

- A) $y=2x+C$
- B) $y=Ce^{(2x)}$
- C) $y=Ce^x$
- D) $y=2e^x+C$

Q14. [MCQ] The distance between points (1,2,3) and (4,6,3) is:

- A) 3
- B) 4
- C) 5
- D) 6

Q15. [MCQ] If vectors a and b have magnitudes 2 and 3 and angle 60° , $|a \times b|$ equals:

- A) $3\sqrt{3}/2$
- B) $3\sqrt{3}$
- C) 6
- D) $9/2$

Q16. [MCQ] The number of solutions of $\sin x = x/3$ for $x \in \mathbb{R}$ is:

- A) 0
- B) 1
- C) 2
- D) 3

Q17. [MCQ] If the roots of $x^2 - px + q = 0$ are positive and reciprocal, then:

- A) $p=q$
- B) $q=1$
- C) $p=1$
- D) $p+q=1$

Q18. [MCQ] The number of distinct arrangements of the letters of MISSISSIPPI is:

- A) 34650
- B) 69300
- C) 11550
- D) 3465

Q19. [MCQ] Two fair dice are thrown. Probability that the sum is prime is:

- A) $5/12$
- B) $7/18$
- C) $1/2$
- D) $13/36$

Q20. [MCQ] If a, b, c are in GP and $a+c=10$, $b=4$, then ac is:

- A) 8
- B) 12
- C) 16
- D) 20

Q21. [NUM] Find the sum of the first 20 terms of the AP 7, 11, 15,...

Enter the required numerical value.

Q22. [NUM] If $|z|=5$ and z has argument $\pi/3$, find $\operatorname{Re}(z)$.

Enter the required numerical value.

Q23. [NUM] Find the number of integer solutions of $|2x-3|<7$.

Enter the required numerical value.

Q24. [NUM] The determinant $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix}$ equals:

Enter the required numerical value.

Q25. [NUM] If $\tan\theta=3/4$ and θ is acute, find $25 \sin^2\theta$.

Enter the required numerical value.

FINAL ANSWER KEY

PHYSICS

1. C	2. A	3. C	4. A	5. B
6. A	7. D	8. C	9. C	10. B
11. C	12. C	13. B	14. C	15. C
16. D	17. B	18. B	19. C	20. B
21. 25	22. $\sqrt{55}$	23. 10	24. 60	25. 8

CHEMISTRY

1. B	2. B	3. B	4. C	5. B
6. B	7. B	8. B	9. B	10. B
11. C	12. B	13. B	14. B	15. C
16. C	17. B	18. B	19. B	20. C
21. 1247.1	22. 2.236	23. 10	24. 3	25. 2

MATHEMATICS

1. A	2. B	3. C	4. C	5. B
6. A	7. B	8. A	9. B	10. B
11. B	12. B	13. B	14. C	15. B
16. D	17. B	18. A	19. A	20. C
21. 900	22. 2.5	23. 6	24. 1	25. 9

DETAILED SOLUTIONS

PHYSICS

Q1 — Correct Answer: C

Concept Used: Kinematics

Step-by-Step Solution: $v=3(t-1)(t-3)$. Turning points are $t=1,3$. Positions: $x(0)=0$, $x(1)=4$, $x(3)=0$, $x(4)=4$. Distance $=4+4+4=12$; displacement $=4$. Ratio $=3$.

Final Answer: C

Q2 — Correct Answer: A

Concept Used: Non-inertial frame

Step-by-Step Solution: In the wedge frame, pseudo-force ma acts opposite wedge acceleration. For zero relative acceleration along the incline: $ma \cos\theta = mg \sin\theta$, hence $a=g \tan\theta$.

Final Answer: A

Q3 — Correct Answer: C

Concept Used: Rotation + energy

Step-by-Step Solution: CM falls by $L/2$. $MgL/2=(1/2)(ML^2/3)\omega^2$, giving $\omega^2=3g/L$.

Final Answer: C

Q4 — Correct Answer: A

Concept Used: Energy

Step-by-Step Solution: Drop in height is R , so kinetic energy at bottom is mgR . Friction μmg over distance d removes mgR , hence $d=R/\mu$.

Final Answer: A

Q5 — Correct Answer: B

Concept Used: Newton's laws

Step-by-Step Solution: Equations: $T-4=2a$ and $10-T=a$. Adding gives $6=3a$, hence $a=2 \text{ m/s}^2$.

Final Answer: B

Q6 — Correct Answer: A

Concept Used: Angular momentum

Step-by-Step Solution: For circular initial orbit $L=mvr$. With L unchanged and $r'=4r$, $v'=L/(mr')=v/4$.

Final Answer: A

Q7 — Correct Answer: D

Concept Used: Surface tension

Step-by-Step Solution: $h=2T \cos\theta/(\rho g r)$. Reducing g by factor 4 increases h by factor 4.

Final Answer: D

Q8 — Correct Answer: C

Concept Used: Thermodynamics

Step-by-Step Solution: $P'=PV^0$ relation: $P(V)=PV^0$? With $P=kV^2$, $k=P/V^2$. $W=\int V^{\wedge}\{2V\} kV^2 dV=(P/V^2)[(8V^3-V^3)/3]=7PV/3$.

Final Answer: C

Q9 — Correct Answer: C

Concept Used: SHM energy

Step-by-Step Solution: $U/E=x^2/A^2=1/4$, so $K/E=3/4$. Thus $K/U=3$.

Final Answer: C

Q10 — Correct Answer: B

Concept Used: Waves

Step-by-Step Solution: $f \propto \sqrt{T/L}$. Factor $= \sqrt{1.44/0.8} = 1.2/0.8 = 1.5$, so correct is B.

Final Answer: B

Q11 — Correct Answer: C

Concept Used: Electrostatics

Step-by-Step Solution: Electric field inside conductor must vanish, so induced inner charge is $-q$. The outer surface then carries $Q+q$.

Final Answer: C

Q12 — Correct Answer: C

Concept Used: Capacitors

Step-by-Step Solution: Total charge $= CV + 2CV = 3CV$; total capacitance $= 3C$, so $V_{\text{final}} = V$.

Final Answer: C

Q13 — Correct Answer: B

Concept Used: Current electricity

Step-by-Step Solution: Maximum power occurs at $R=r$. Efficiency $= R/(R+r) = 1/2$.

Final Answer: B

Q14 — Correct Answer: C

Concept Used: Magnetic motion

Step-by-Step Solution: $r = mv/qB$ and $K \propto v^2$, hence $r \propto \sqrt{K}$. Doubling K gives $\sqrt{2}r$.

Final Answer: C

Q15 — Correct Answer: C

Concept Used: EMI

Step-by-Step Solution: Small element dx has $d\epsilon = B(\omega x)dx$. Integrating 0 to l gives $\frac{1}{2}B\omega l^2$.

Final Answer: C

Q16 — Correct Answer: D

Concept Used: AC circuits

Step-by-Step Solution: At resonance $X_L = X_C$. At 2ω , X_L doubles while X_C halves, so $|X_L - X_C| > 0$ and $Z = \sqrt{R^2 + \text{positive}} > R$.

Final Answer: D

Q17 — Correct Answer: B

Concept Used: Ray optics

Step-by-Step Solution: First image is 60 cm behind lens. Mirror at +10 cm reflects rays; object for second lens is at 50 cm behind lens, giving a virtual-object calculation that returns the final image 30 cm in front of lens.

Final Answer: B

Q18 — Correct Answer: B

Concept Used: Wave optics

Step-by-Step Solution: $I = 4I_1 \cos^2(\phi/2)$. At $\phi = \pi/3$, $I = 3I_1$. Therefore $I_{\text{max}} = 4I_1 = 4I_0/3$.

Final Answer: B

Q19 — Correct Answer: C

Concept Used: Photoelectric effect

Step-by-Step Solution: $eV_0 = hc/\lambda - \phi$. At $\lambda/2$: $eV' = 2hc/\lambda - \phi = eV_0 + hc/\lambda$.

Final Answer: C

Q20 — Correct Answer: B

Concept Used: Nuclear physics

Step-by-Step Solution: Total BE=1800 MeV. Using $1\text{ u} \approx 931.5\text{ MeV}$, $\Delta m \approx 1.93\text{ u}$.

Final Answer: B

Q21 — Correct Answer: 25

Concept Used: Kinematics

Step-by-Step Solution: Height reached $= u^2/(2g) = 20\text{ m}$ in 2 s; in the next 1 s it falls 5 m. Total distance = 25 m.

Final Answer: 25

Q22 — Correct Answer: $\sqrt{55}$

Concept Used: Work-energy

Step-by-Step Solution: Net work $= \int_0^5 [(10+2x)-4]dx = \int_0^5 (6+2x)dx = 55\text{ J}$. Thus $\frac{1}{2}(2)v^2 = 55$, so $v = \sqrt{55}\text{ m/s}$.

Final Answer: $\sqrt{55}$

Q23 — Correct Answer: 10

Concept Used: Capacitors

Step-by-Step Solution: Charge remains constant. $U' = U/K = (\frac{1}{2}CV^2)/3 = 10\text{ mJ}$.

Final Answer: 10

Q24 — Correct Answer: 60

Concept Used: Chemical kinetics

Step-by-Step Solution: Remaining fraction $= 12.5\% = 1/8 = (1/2)^3$, so 3 half-lives = 60 min.

Final Answer: 60

Q25 — Correct Answer: 8

Concept Used: Quadratic equations

Step-by-Step Solution: If roots α, β , $\alpha + \beta = 6$ and $|\alpha - \beta| = 2$. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = 4 \Rightarrow 36 - 4k = 4 \Rightarrow k = 8$.

Final Answer: 8

CHEMISTRY

Q1 — Correct Answer: B

Concept Used: Thermodynamics

Step-by-Step Solution: $W = nRT \ln(V_2/V_1) = 0.1 \times 8.314 \times 300 \times \ln 4 \approx 0.69\text{ kJ}$.

Final Answer: B

Q2 — Correct Answer: B

Concept Used: Ionic equilibrium

Step-by-Step Solution: Henderson-Hasselbalch: $\text{pH} = \text{pK}_a + \log(1) = \text{pK}_a$.

Final Answer: B

Q3 — Correct Answer: B

Concept Used: Chemical equilibrium

Step-by-Step Solution: Fewer gas moles are on the left (1 vs 2), so pressure increase favors left.

Final Answer: B

Q4 — Correct Answer: C

Concept Used: Electrochemistry

Step-by-Step Solution: $E^\circ_{\text{cell}} = 0.80 - (-0.76) = 1.56\text{ V}$.

Final Answer: C

Q5 — Correct Answer: B

Concept Used: Chemical kinetics

Step-by-Step Solution: $t_{1/2} = 0.693/k \approx 300$ s.

Final Answer: B

Q6 — Correct Answer: B

Concept Used: Chemical bonding

Step-by-Step Solution: Removing an electron from antibonding π^* raises bond order: $O_2 = 2$, $O_2^+ = 2.5$, $O_2^- = 1.5$, $O_2^{2-} = 1$.

Final Answer: B

Q7 — Correct Answer: B

Concept Used: Organic chemistry

Step-by-Step Solution: 2-propanol gives iodoform and contains an O-H bond, reacting with Na.

Final Answer: B

Q8 — Correct Answer: B

Concept Used: Elimination

Step-by-Step Solution: Zaitsev alkene, 2-methyl-2-butene, is more substituted.

Final Answer: B

Q9 — Correct Answer: B

Concept Used: Phenols

Step-by-Step Solution: Phenol gives violet/colored ferric phenolate complex with neutral $FeCl_3$; ethanol does not.

Final Answer: B

Q10 — Correct Answer: B

Concept Used: Reaction mechanism

Step-by-Step Solution: Ionization of substrate to carbocation is the slow step.

Final Answer: B

Q11 — Correct Answer: C

Concept Used: Solutions

Step-by-Step Solution: $\Delta T_f = iK_f m \Rightarrow i = 0.186 / (1.86 \times 0.1) = 1$.

Final Answer: C

Q12 — Correct Answer: B

Concept Used: Solid state

Step-by-Step Solution: 8 corners contribute 1; 6 faces contribute 3; total 4.

Final Answer: B

Q13 — Correct Answer: B

Concept Used: Coordination chemistry

Step-by-Step Solution: $x + 6(-1) = -4 \Rightarrow x = +2$.

Final Answer: B

Q14 — Correct Answer: B

Concept Used: Intermolecular forces

Step-by-Step Solution: Ethanol forms strong intermolecular H-bonds.

Final Answer: B

Q15 — Correct Answer: C

Concept Used: d-block

Step-by-Step Solution: Mn^{2+} is $3d^5$, hence five unpaired electrons.

Final Answer: C

Q16 — Correct Answer: C

Concept Used: p-block

Step-by-Step Solution: Al_2O_3 reacts with both acids and bases.

Final Answer: C

Q17 — Correct Answer: B

Concept Used: Chemical kinetics

Step-by-Step Solution: $\text{Rate} \propto [\text{A}]^1[\text{B}]^0$, so order = 1.

Final Answer: B

Q18 — Correct Answer: B

Concept Used: Isomerism

Step-by-Step Solution: Each alkene carbon in but-2-ene has two different substituents.

Final Answer: B

Q19 — Correct Answer: B

Concept Used: Coordination chemistry

Step-by-Step Solution: Ag^+ forms $[\text{Ag}(\text{NH}_3)_2]^+$, lowering free Ag^+ concentration.

Final Answer: B

Q20 — Correct Answer: C

Concept Used: Kinetic theory

Step-by-Step Solution: Ideal gas obeys $PV = nRT$; intermolecular forces are neglected and collisions are elastic.

Final Answer: C

Q21 — Correct Answer: 1247.1

Concept Used: Thermodynamics

Step-by-Step Solution: $\Delta U = nC_v\Delta T = 2 \times (3R/2) \times 50 = 150R = 1247.1 \text{ J}$.

Final Answer: 1247.1

Q22 — Correct Answer: 2.236

Concept Used: Ionic equilibrium

Step-by-Step Solution: $[\text{H}^+] = \sqrt{K_a C} = \sqrt{(5 \times 10^{-6})} = 2.236 \times 10^{-3} \text{ M}$.

Final Answer: 2.236

Q23 — Correct Answer: 10

Concept Used: Chemical kinetics

Step-by-Step Solution: 25% remains $= 1/4 = (1/2)^2$, so 20 min = 2 half-lives; $t_{1/2} = 10 \text{ min}$.

Final Answer: 10

Q24 — Correct Answer: 3

Concept Used: Ionic equilibrium

Step-by-Step Solution: Strong acid fully dissociates: $[\text{H}^+] = 10^{-3} \text{ M}$, $\text{pH} = 3$.

Final Answer: 3

Q25 — Correct Answer: 2

Concept Used: Coordination chemistry

Step-by-Step Solution: One Cl^- is coordinated; two Cl^- are counter-ions outside the sphere.

Final Answer: 2

MATHEMATICS

Q1 — Correct Answer: A

Concept Used: Complex numbers

Step-by-Step Solution: $z=i$, so $z^8=i^8=1$.

Final Answer: A

Q2 — Correct Answer: B

Concept Used: Inequality

Step-by-Step Solution: $x^2+4/x^2 \geq 2\sqrt{4}=4$.

Final Answer: B

Q3 — Correct Answer: C

Concept Used: Modulus

Step-by-Step Solution: For $x \geq 1$, $2x=4 \rightarrow x=2$; for $x \leq -1$, $-2x=4 \rightarrow x=-2$.

Final Answer: C

Q4 — Correct Answer: C

Concept Used: Matrices

Step-by-Step Solution: For 2×2 , $\det(kA)=k^2 \det A=4 \times 3=12$.

Final Answer: C

Q5 — Correct Answer: B

Concept Used: Binomial theorem

Step-by-Step Solution: $(1+x)^4(1-x)^4=(1-x^2)^4$ contains only even powers, so x^3 coefficient is 0.

Final Answer: B

Q6 — Correct Answer: A

Concept Used: Circle

Step-by-Step Solution: For $x^2+y^2=a^2$, tangent at (x_1, y_1) : $xx_1+yy_1=a^2$.

Final Answer: A

Q7 — Correct Answer: B

Concept Used: Conic sections

Step-by-Step Solution: $c^2=25-9=16$, $c=4$; $e=c/a=4/5$.

Final Answer: B

Q8 — Correct Answer: A

Concept Used: Limits

Step-by-Step Solution: $\sin 3x=3x-27x^3/6$; $3\sin x=3x-3x^3/6$; difference $=-24x^3/6=-4x^3$.

Final Answer: A

Q9 — Correct Answer: B

Concept Used: Continuity

Step-by-Step Solution: Left derivative at 0 is -1 and right derivative is $+1$.

Final Answer: B

Q10 — Correct Answer: B

Concept Used: AOD

Step-by-Step Solution: Parabola has vertex $x=2$, value 4.

Final Answer: B

Q11 — Correct Answer: B

Concept Used: Integration

Step-by-Step Solution: $\text{Integral}=\tan^{-1}x|_0^1=\pi/4$.

Final Answer: B

Q12 — Correct Answer: B

Concept Used: Definite integration

Step-by-Step Solution: $I_n = (n-1)/n I_{n-2}$; equality $I_n = I_{n+2}$ gives $n=1$.

Final Answer: B

Q13 — Correct Answer: B

Concept Used: Differential equations

Step-by-Step Solution: $dy/y = 2dx$; $\ln|y| = 2x + C$.

Final Answer: B

Q14 — Correct Answer: C

Concept Used: 3D geometry

Step-by-Step Solution: Distance $= \sqrt{3^2 + 4^2} = 5$.

Final Answer: C

Q15 — Correct Answer: B

Concept Used: Vector algebra

Step-by-Step Solution: $|a \times b| = ab \sin 60 = 6 \times \sqrt{3}/2 = 3\sqrt{3}$.

Final Answer: B

Q16 — Correct Answer: D

Concept Used: Trigonometric equations

Step-by-Step Solution: $\sin x = x/3$ has $x=0$ and one positive solution in $(0, \pi)$, plus its negative by odd symmetry: 3 solutions.

Final Answer: D

Q17 — Correct Answer: B

Concept Used: Quadratic equations

Step-by-Step Solution: Product of reciprocal roots is 1, so $q=1$.

Final Answer: B

Q18 — Correct Answer: A

Concept Used: Permutation

Step-by-Step Solution: $11!/(4!4!2!) = 34650$.

Final Answer: A

Q19 — Correct Answer: A

Concept Used: Probability

Step-by-Step Solution: Prime sums 2,3,5,7,11 have counts $1+2+4+6+2=15$; $15/36=5/12$.

Final Answer: A

Q20 — Correct Answer: C

Concept Used: Progressions

Step-by-Step Solution: $b^2 = ac = 16$.

Final Answer: C

Q21 — Correct Answer: 900

Concept Used: Progressions

Step-by-Step Solution: $S_{20} = 20/2 [14 + 19 \times 4] = 10 \times 90 = 900$.

Final Answer: 900

Q22 — Correct Answer: 2.5

Concept Used: Complex numbers

Step-by-Step Solution: $\operatorname{Re}(z) = |z|\cos\pi/3 = 5/2$.

Final Answer: 2.5

Q23 — Correct Answer: 6

Concept Used: Inequalities

Step-by-Step Solution: $-7 < 2x - 3 < 7 \Rightarrow -2 < x < 5$; actually $-4 < 2x < 10 \Rightarrow -2 < x < 5$, integers $-1, 0, 1, 2, 3, 4$: 6.

Final Answer: 6

Q24 — Correct Answer: 1

Concept Used: Determinants

Step-by-Step Solution: Expansion gives $1(-24) - 2(-20) + 3(-5) = 1$.

Final Answer: 1

Q25 — Correct Answer: 9

Concept Used: Trigonometry

Step-by-Step Solution: $\sin\theta = 3/5$, so $25\sin^2\theta = 9$.

Final Answer: 9

FINAL QUALITY-AUDIT NOTE

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- Structure audited for 75 total questions: 25 Physics + 25 Chemistry + 25 Mathematics.
- All single-correct questions have one designated answer; numerical questions have a stated numerical result.
- Calculations and answer-key consistency were checked during construction, and malformed draft items were replaced before PDF generation.
- Questions are original formulations using established JEE concepts and are not presented as official previous-year questions.
- Difficulty is intentionally skewed toward difficult/very difficult conceptual application rather than raw arithmetic length.